APS-ES-Series
1500 V_{DC}

Outdoor Energy Storage hybrid inverters, with output power range from 600 kVA up to 5000 kVA.



The APS-Series 1500 V_{DC} outdoor central inverter is the most cost-effective solution for large-scale Energy Storage installations.

The APS is a turnkey solution made for the most demanding environments.

- **Ready for 1500 V_{DC} systems –** Breakthrough technology for the newest industry standards.
- **IP65 outdoor cabinet –** Hermetically sealed against heavy dust, sand and rain.
- **Two redundant cooling systems** ensuring energy production up to 60 °C ambient.
- **Low maintenance heat exchangers –** Easy to clean air-to-air heat exchanger with no filter mats.
- Internal maintenance free **patented air dehumidifier** to avoid condensation.
- **AC circuit breaker** built-in.
- **Optional fuse-protected DC inputs**.
- **Modbus TCP or optional Real-Time Ethernet** communication interfaces.
- **Optional and configurable hybrid system** for storage applications and STATCOM function in the same inverter.

APS-ES-Series 1500 V_{DC}

600 … 5000 kVA

DATASHEET - INTRODUCTION

Models	**APS600-ES-1-330-5** **APS650-ES-1-360-5** **APS725-ES-1-400-5** **APS800-ES-1-440-5** **APS870-ES-1-480-5** **APS945-ES-1-520-5** **APS1000-ES-1-550-5** **APS1045-ES-1-575-5** **APS1090-ES-1-600-5** **APS1140-ES-1-630-5** **APS1200-ES-1-660-5** **APS1250-ES-1-690-5**	**APS1200-ES-2-330-5** **APS1300-ES-2-360-5** **APS1450-ES-2-400-5** **APS1600-ES-2-440-5** **APS1740-ES-2-480-5** **APS1890-ES-2-520-5** **APS2000-ES-2-550-5** **APS2090-ES-2-575-5** **APS2180-ES-2-600-5** **APS2280-ES-2-630-5** **APS2400-ES-2-660-5** **APS2500-ES-2-690-5**	**APS1800-ES-3-330-5** **APS1950-ES-3-360-5** **APS2175-ES-3-400-5** **APS2400-ES-3-440-5** **APS2610-ES-3-480-5** **APS2835-ES-3-520-5** **APS3000-ES-3-550-5** **APS3135-ES-3-575-5** **APS3270-ES-3-600-5** **APS3420-ES-3-630-5** **APS3600-ES-3-660-5** **APS3750-ES-3-690-5**	**APS2400-ES-4-330-5** **APS2600-ES-4-360-5** **APS2900-ES-4-400-5** **APS3200-ES-4-440-5** **APS3480-ES-4-480-5** **APS3780-ES-4-520-5** **APS4000-ES-4-550-5** **APS4180-ES-4-575-5** **APS4360-ES-4-600-5** **APS4560-ES-4-630-5** **APS4800-ES-4-660-5** **APS5000-ES-4-690-5**
Cabinet options				
Number of Apparent Power Units (APU)	1	2	3	4
Nominal grid voltage ($U_{AC, nom}$) options	330 V / 360 V / 400 V / 440 V / 480 V / 520 V / 550 V / 575 V / 600 V / 630 V / 660 V / 690 V			
Max. AC apparent power (S_{max}) options	From 600 kVA up to 2500 kVA		From 1800 kVA up to 5000 kVA	
Max. DC voltage ($U_{DC, max}$)	1500 V			

APS-ES-Series 1500 V_{DC} – 330 V_{AC} 600 … 2400 kVA

(I) TECHNICAL DATA

	APS600-ES-1-330-5	APS1200-ES-2-330-5	APS1800-ES-3-330-5	APS2400-ES-4-330-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	600 kVA	1200 kVA	1800 kVA	2400 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	600 kW	1200 kW	1800 kW	2400 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	330 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	100 kVA	200 kVA	300 kVA	400 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 330 V_{AC} 600 … 2400 kVA

(II) TECHNICAL DATA

	APS600-ES-1-330-5	APS1200-ES-2-330-5	APS1800-ES-3-330-5	APS2400-ES-4-330-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	483 V / 531 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	610 kW	1220 kW	1829 kW	2439 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 360 V_{AC} 650 … 2600 kVA

(I) TECHNICAL DATA

	APS650-ES-1-360-5	APS1300-ES-2-360-5	APS1950-ES-3-360-5	APS2600-ES-4-360-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	650 kVA	1300 kVA	1950 kVA	2600 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	650 kW	1300 kW	1950 kW	2600 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	360 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	108 kVA	217 kVA	325 kVA	433 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 360 V_{AC} 650 … 2600 kVA

(II) TECHNICAL DATA

	APS650-ES-1-360-5	APS1300-ES-2-360-5	APS1950-ES-3-360-5	APS2600-ES-4-360-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	527 V / 580 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	661 kW	1321 kW	1982 kW	2642 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	(97.5 | 98.0 | 98.0 | 97.9 | 97.8) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	(97.5 | 98.0 | 98.0 | 97.8 | 97.7) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 400 V_{AC} 725 … 2900 kVA

(I) TECHNICAL DATA

	APS725-ES-1-400-5	APS1450-ES-2-400-5	APS2175-ES-3-400-5	APS2900-ES-4-400-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	725 kVA	1450 kVA	2175 kVA	2900 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	725 kW	1450 kW	2175 kW	2900 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	400 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	121 kVA	242 kVA	363 kVA	483 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 400 V_{AC} 725 … 2900 kVA

(II) TECHNICAL DATA

	APS725-ES-1-400-5	APS1450-ES-2-400-5	APS2175-ES-3-400-5	APS2900-ES-4-400-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	585 V / 644 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	737 kW	1474 kW	2210 kW	2947 kW	
Max. capacity against earthing	2000 µF				For each IT system

GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 440 V_{AC} 800 … 3200 kVA

(I) TECHNICAL DATA

	APS800-ES-1-440-5	APS1600-ES-2-440-5	APS2400-ES-3-440-5	APS3200-ES-4-440-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	800 kVA	1600 kVA	2400 kVA	3200 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	800 kW	1600 kW	2400 kW	3200 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	440 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	133 kVA	267 kVA	400 kVA	533 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 440 V_{AC}

800 … 3200 kVA

(II) TECHNICAL DATA

	APS800-ES-1-440-5	APS1600-ES-2-440-5	APS2400-ES-3-440-5	APS3200-ES-4-440-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	643 V / 707 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	813 kW	1626 kW	2439 kW	3252 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 480 V_{AC} 870 … 3480 kVA

(I) TECHNICAL DATA

	APS870-ES-1-480-5	APS1740-ES-2-480-5	APS2610-ES-3-480-5	APS3480-ES-4-480-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	870 kVA	1740 kVA	2610 kVA	3480 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	870 kW	1740 kW	2610 kW	3480 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	480 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	145 kVA	290 kVA	435 kVA	580 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 480 V_{AC} 870 … 3480 kVA

(II) TECHNICAL DATA

	APS870-ES-1-480-5	APS1740-ES-2-480-5	APS2610-ES-3-480-5	APS3480-ES-4-480-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	702 V / 772 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	884 kW	1768 kW	2652 kW	3537 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	(97.6 | 98.1 | 98.1 | 98.0 | 97.9) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	(97.6 | 98.1 | 98.1 | 97.9 | 97.8) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 520 V_{AC} 945 … 3780 kVA

(I) TECHNICAL DATA

	APS945-ES-1-520-5	APS1890-ES-2-520-5	APS2835-ES-3-520-5	APS3780-ES-4-520-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	945 kVA	1890 kVA	2835 kVA	3780 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	945 kW	1890 kW	2835 kW	3780 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	520 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	158 kVA	315 kVA	473 kVA	630 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 520 V_{AC} 945 … 3780 kVA

(II) TECHNICAL DATA

	APS945-ES-1-520-5	APS1890-ES-2-520-5	APS2835-ES-3-520-5	APS3780-ES-4-520-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	760 V / 836 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	960 kW	1921 kW	2881 kW	3841 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.2 | 98.6 | 98.6 | 98.5 | 98.3) %	(98.2 | 98.6 | 98.6 | 98.5 | 98.3) %	(98.2 | 98.6 | 98.6 | 98.5 | 98.3) %	(98.2 | 98.6 | 98.6 | 98.5 | 98.3) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.2 | 98.6 | 98.6 | 98.4 | 98.2) %	(98.2 | 98.6 | 98.6 | 98.4 | 98.2) %	(98.2 | 98.6 | 98.6 | 98.4 | 98.2) %	(98.2 | 98.6 | 98.6 | 98.4 | 98.2) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.5%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 550 V_{AC} 1000 … 4000 kVA

(I) TECHNICAL DATA

	APS1000-ES-1-550-5	APS2000-ES-2-550-5	APS3000-ES-3-550-5	APS4000-ES-4-550-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1000 kVA	2000 kVA	3000 kVA	4000 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1000 kW	2000 kW	3000 kW	4000 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	550 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	167 kVA	333 kVA	500 kVA	667 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 550 V_{AC} 1000 … 4000 kVA

(II) TECHNICAL DATA

	APS1000-ES-1-550-5	APS2000-ES-2-550-5	APS3000-ES-3-550-5	APS4000-ES-4-550-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	804 V / 884 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1016 kW	2033 kW	3049 kW	4065 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.3 | 98.7 | 98.7 | 98.6 | 98.4) %	(98.3 | 98.7 | 98.7 | 98.6 | 98.4) %	(98.3 | 98.7 | 98.7 | 98.6 | 98.4) %	(98.3 | 98.7 | 98.7 | 98.6 | 98.4) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.3 | 98.7 | 98.7 | 98.5 | 98.3) %	(98.3 | 98.7 | 98.7 | 98.5 | 98.3) %	(98.3 | 98.7 | 98.7 | 98.5 | 98.3) %	(98.3 | 98.7 | 98.7 | 98.5 | 98.3) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.6%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Maximum auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 575 V_{AC} 1045 … 4180 kVA

(I) TECHNICAL DATA

	APS1045-ES-1-575-5	APS2090-ES-2-575-5	APS3135-ES-3-575-5	APS4180-ES-4-575-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1045 kVA	2090 kVA	3135 kVA	4180 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1045 kW	2090 kW	3135 kW	4180 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	575 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	174 kVA	348 kVA	523 kVA	697 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 575 V_{AC} 1045 … 4180 kVA

(II) TECHNICAL DATA

	APS1045-ES-1-575-5	APS2090-ES-2-575-5	APS3135-ES-3-575-5	APS4180-ES-4-575-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	840 V / 924 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1062 kW	2124 kW	3186 kW	4248 kW	
Max. capacity against earthing	2000 µF				For each IT system

GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.3 | 98.8 | 98.8 | 98.6 | 98.5) %	(98.3 | 98.8 | 98.8 | 98.6 | 98.5) %	(98.3 | 98.8 | 98.8 | 98.6 | 98.5) %	(98.3 | 98.8 | 98.8 | 98.6 | 98.5) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.3 | 98.8 | 98.8 | 98.5 | 98.4) %	(98.3 | 98.8 | 98.8 | 98.5 | 98.4) %	(98.3 | 98.8 | 98.8 | 98.5 | 98.4) %	(98.3 | 98.8 | 98.8 | 98.5 | 98.4) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.7%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Maximum auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 600 V_{AC} 1090 … 4360 kVA

(I) TECHNICAL DATA

	APS1090-ES-1-600-5	APS2180-ES-2-600-5	APS3270-ES-3-600-5	APS4360-ES-4-600-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1090 kVA	2180 kVA	3270 kVA	4360 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1090 kW	2180 kW	3270 kW	4360 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	600 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	182 kVA	363 kVA	545 kVA	727 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 600 V_{AC} 1090 … 4360 kVA

(II) TECHNICAL DATA

	APS1090-ES-1-600-5	APS2180-ES-2-600-5	APS3270-ES-3-600-5	APS4360-ES-4-600-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	877 V / 965 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1108 kW	2215 kW	3323 kW	4431 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.4 | 98.9 | 98.8 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.8 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.8 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.8 | 98.7 | 98.5) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.4 | 98.8 | 98.8 | 98.6 | 98.4) %	(98.4 | 98.8 | 98.8 | 98.6 | 98.4) %	(98.4 | 98.8 | 98.8 | 98.6 | 98.4) %	(98.4 | 98.8 | 98.8 | 98.6 | 98.4) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.7%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 630 V_{AC} 1140 … 4560 kVA

(I) TECHNICAL DATA

	APS1140-ES-1-630-5	APS2280-ES-2-630-5	APS3420-ES-3-630-5	APS4560-ES-4-630-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1140 kVA	2280 kVA	3420 kVA	4560 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1140 kW	2280 kW	3420 kW	4560 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	630 V				3~, phase to phase
Grid voltage range	+/- 10% $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	190 kVA	380 kVA	570 kVA	760 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 630 V_{AC} 1140 … 4560 kVA

(II) TECHNICAL DATA

	APS1140-ES-1-630-5	APS2280-ES-2-630-5	APS3420-ES-3-630-5	APS4560-ES-4-630-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1000 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	920 V / 1012 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1159 kW	2317 kW	3476 kW	4634 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.4 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.4 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.4 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.4 | 98.9 | 98.9 | 98.8 | 98.6) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.4 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.4 | 98.9 | 98.9 | 98.7 | 98.5) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.8%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 660 V_{AC} 1200 … 4800 kVA

(I) TECHNICAL DATA

	APS1200-ES-1-660-5	APS2400-ES-2-660-5	APS3600-ES-3-660-5	APS4800-ES-4-660-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1200 kVA	2400 kVA	3600 kVA	4800 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1200 kW	2400 kW	3600 kW	4800 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	660 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	200 kVA	400 kVA	600 kVA	800 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 660 V_{AC} 1200 … 4800 kVA

(II) TECHNICAL DATA

	APS1200-ES-1-660-5	APS2400-ES-2-660-5	APS3600-ES-3-660-5	APS4800-ES-4-660-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1100 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	964 V / 1060 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1220 kW	2439 kW	3659 kW	4878 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.5 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.5 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.5 | 98.9 | 98.9 | 98.8 | 98.6) %	(98.5 | 98.9 | 98.9 | 98.8 | 98.6) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.5 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.5 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.5 | 98.9 | 98.9 | 98.7 | 98.5) %	(98.5 | 98.9 | 98.9 | 98.7 | 98.5) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.8%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 690 V_{AC} 1250 … 5000 kVA

(I) TECHNICAL DATA

	APS1250-ES-1-690-5	APS2500-ES-2-690-5	APS3750-ES-3-690-5	APS5000-ES-4-690-5	COMMENTS
GRID SIDE					
Max. AC apparent power (S_{max})	1250 kVA	2500 kVA	3750 kVA	5000 kVA	At nominal grid voltage
Nominal AC power ($P_{AC, nom}$)	1250 kW	2500 kW	3750 kW	5000 kW	At (cos ϕ) = 1.0
Number of Apparent Power Units (APU)	1	2	3	4	
Number of independent grids	1		2		
Nominal grid voltage ($U_{AC, nom}$)	690 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom}$				
Nominal grid frequency (f_{nom})	50 Hz				60 Hz option available
RoCoF	<= 4Hz/s				Other on request
Network configuration	IT system				
Max. AC current per APU ($I_{AC, max\ (APU)}$)	1050 A				
Max. AC current – APS ($I_{AC, max\ (APS)}$)	1050 A	2 x 1050 A	3 x 1050 A	4 x 1050 A	
Max. short circuit level ($I_{AC, SC\ (APS)}$)	42 kA				
Short circuit withstand capability (I_{CW})	42 kA, 100ms				
Short circuit contribution (I'_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I''_K)	1060 A	2 x 1060 A	3 x 1060 A	4 x 1060 A	Max. RMS value
Short circuit contribution (I_P)	1560 A	2 x 1560 A	3 x 1560 A	4 x 1560 A	Max. Peak value
Max. load capability	100%				At Operating mode Island grid
Time span overload	0.0 s				At Operating mode Island grid
Asymmetrical load between the phases	208 kVA	417 kVA	625 kVA	833 kVA	At Operating mode Island grid
Power factor (cos ϕ)	0 ind … 1 … 0 cap				Adjustable
AC current distortion (THD)	< 3%				

APS-ES-Series 1500 V_{DC} – 690 V_{AC} 1250 … 5000 kVA

(II) TECHNICAL DATA

	APS1250-ES-1-690-5	APS2500-ES-2-690-5	APS3750-ES-3-690-5	APS5000-ES-4-690-5	COMMENTS
DC SIDE					
Independent DC sources	1		2		Depending on configuration
Nominal DC voltage ($U_{DC, nom}$)	1100 V				
Max. DC voltage ($U_{DC, max}$)	1500 V				Depending on application
Min. DC voltage ($U_{DC, min}$)	1008 V / 1109 V				At 100% / 110% nominal grid voltage (*)
Max. DC current ($I_{DC, max}$)	1220 A	2 x 1220 A	3 x 1220 A	4 x 1220 A	
Max. short circuit level ($I_{DC, SC (APS)}$)	6.4 kA / 30 kA / 140 kA (max. 0.5 ms)				Without / With DC fuses / External protection required
Nominal DC power ($P_{DC, nom}$)	1270 kW	2541 kW	3811 kW	5081 kW	
Max. capacity against earthing	2000 µF				For each IT system
GENERAL					
Control strategy	CC-CV				
Efficiency (coupled APUs)	(98.5 | 99.0 | 98.9 | 98.8 | 98.7) %	(98.5 | 99.0 | 98.9 | 98.8 | 98.7) %	(98.5 | 99.0 | 98.9 | 98.8 | 98.7) %	(98.5 | 99.0 | 98.9 | 98.8 | 98.7) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
Efficiency (separated APUs)	(98.5 | 99.0 | 98.9 | 98.7 | 98.6) %	(98.5 | 99.0 | 98.9 | 98.7 | 98.6) %	(98.5 | 99.0 | 98.9 | 98.7 | 98.6) %	(98.5 | 99.0 | 98.9 | 98.7 | 98.6) %	At (10 | 30 | 50 | 75 | 100) % power, at $U_{DC, nom}$
EU efficiency	98.8%				Including all inverter losses
Feed-in starting at ($P_{DC, th}$)	200 W	200 W	400 W	400 W	
Standby losses	80 W	90 W	140 W	150 W	
Max. auxiliary power	< 3500 W		< 7000 W		

(*) for further information please have a look
Pac(Udc)➔ Datasheet page 36 ff
Sac(Udc)➔ Additional technical data document

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(III) TECHNICAL DATA

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	COMMENTS
AUXILIARY SUPPLY (EXTERNAL)					
Nominal grid voltage ($U_{AC, nom (aux)}$)	400 V				3~, phase to phase
Grid voltage range	+/- 10% of $U_{AC, nom (aux)}$				
Nominal grid frequency ($f_{nom (aux)}$)	50 Hz				
Network configuration	TN-S system				
Max. AC current ($I_{AC, (aux)}$)	3 x C13A				
Max. short circuit level ($I_{AC, SC (APS)}$)	6 kA				
Internal buffer time	5.0 s				Only for control supply available

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(I) GENERAL DATA

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	COMMENTS
PARAMETER					
Ambient working temperature	-40°C / -25°C / -10°C … 60°C				incl. heating + therm. iso / incl. heating / standard
Ambient storage temperature	-25°C … 40°C				
Storage relative humidity	< 80%				Non-condensing
Maximum altitude	1500 m above sea level				Without power derating
Cooling type	3605 m³/h		7210 m³/h		Air flow air-air cooler
	8280 m³/h		16560 m³/h		Air flow water-air cooler
Degree of protection	IP65				Only when AC connection cover is installed with IP54
Dimensions (L × W × H)	2120 × 1170 × 3760		3690 × 1170 × 3760		Dimensions in millimeters
Weight	< 1525 kg	< 2150 kg	< 3275 kg	< 3900 kg	
Shelter surface	Painted				
Corrosivity category	C4-high				Others on request
Colour	RAL7035				Others on request
Snow load	1,5 kN/m²				
Max. permissible wind speed	70 m/s				

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(II) GENERAL DATA

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	COMMENTS
EMC and security standards	IEC 61000-6-2, IEC 61000-6-4 + AMD1, IEC 62109-1, IEC 62109-2.				
CE-conformity	Complies				
Grid connection standards	VDE-AR-N-4110, VDE-AR-N-4120, VDE-AR-N-4130 (Germany), Arrêté du 9 juin 2020+AMD sept 2020+AMD mai 2021+AMD nov 2021 (France),C10/11:2019 + AMD 2021 (Belgium),CEI 0-16: 2022-03 + V1:2022-11+V2:2023-05, Allegato A.68 Rev. 04-Maggio 2022(Italy), Norma Técnica de Supervisión UE 2016/631. Revisión 2.0 (Spain), Engineering Recommendation G99, Issue 1-Amendment 9, October 2022 (UK), EirGrid Grid Code Version 11 (Ireland), Engineering Recommendation G99/NI, Issue 1 April 2019 (Northern Ireland), Vietnam Grid Code, IEC 62116, SJV2019 (Finland), NC RfG and Netcode elektriciteit version 1.4, Netcode elektriciteit (Storage) version 1.1 (Netherland), ORDER 30/2013 (Romania), Grid Connection Criteria of the Solar Photovoltaic and Wind Generation Facilities, Annex 18 (Turkey), TOR Erzeuger V1.2 Type B,C and D – 2022 OVE R25 Procedure (Austria), EIFS 2018:2 (Sweden), EN 50549-1:2019, EN 50549-2:2019				
Efficiency standard	IEC 61683				
UL / CSA standards	UL 1741 second edition Rev. September 16, 2020 with IEEE1547:2018, IEEE1547.1: 2020 and SA (SA8, SA9, SA10, SA11, SA12, SA13, SA14, SA15. SA17 and SA18 with SRD Electric Rule No 21 phase 2 and 3 and Hawaiian Rule 14H SRD V1.1) and NFPA70. CSA C22.2 No. 107.1-16 (Reaffirmed 2016) Compliance with CEC Guideline for use of the Performance Test Protocol for Evaluating Inverters Used in Grid-Connected Photovoltaic Systems				To be ordered as an option
Seismic standards	IEEE 693-2005, EN60068-3-3:1993, EN 60068-2-6:2008, EN 60068-2-47:2005				To be ordered as an option

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(I) FEATURES & OPTIONS – APS - ENERGY STORAGE APPLICATION

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	COMMENTS
IP65* outdoor cabinet made for desert installations	F				* except exterior cooling unit
Insulation monitor	F				
Monitored surge arresters on AC side	F				
Status display on the shelter	O				
Air dehumidifier inside the shelter	F				
External heat exchanger for dusty/moist ambient and extreme climatic conditions	F				
Low-Voltage Ride Through (LVRT) handling	F				
Modbus TCP communication interface	F				
Powerlink communication interface (Real-time Ethernet)	O				
Internal data logger	F				
Service access via VNC	F				
FTP server for log log data	F				

(O) – Optional (F) – Feature

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(II) FEATURES & OPTIONS – APS - ENERGY STORAGE APPLICATION

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	**COMMENTS**
Network Interface	F *)				Network socket
	O *)				Fiber optic multi-mode
	O *)				Fiber optic single-mode
Heating	O				Limited to 40 degrees
Alternative grid voltage range (+/- xx %)	O				Only with external control voltage supply
Grid frequency 60 Hz	O				
Power limitation [kVA]	O				
Area for customer's installations (H x W X D) 400 x 500 x 200 mm	F				Including: 1-phase, 230 V, TN, 1000 VA, network connection
UL certified	O				
Seismic IEEE 693	O				
Island grid functionality	O				Refer to P(Q) diagram for $U_{DC, min}$ requirement

(O) – Optional (F) – Feature (-) – not available (*)) – Only one option possible

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(III) FEATURES & OPTIONS – APS - ENERGY STORAGE APPLICATION

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	**COMMENTS**
Additional auxiliary output	-		F *)		None
	-		O *)		<u>No transformer:</u> fuse-protected output at APU 3/4 with 63 A (3~, PE)
	-		O *)		<u>With transformer:</u> with internal 8 kVA transformer (3~, N, PE)
AC short circuit level 50 kA	-				
MV transformer cooling interface	O				
Overvoltage protection AC side	F/o				Class 2 / Class 1+2
Overvoltage protection DC side	-/F				Class 2 / Class 1+2
Additional overvoltage protection	F				None (only at output of the internal auxiliary transformer
	O				At control interface
	O				At heating interface
Water connection	F				Hose connection
	O				Non-spill connection

(O) – Optional (F) – Feature (-) – not available (*)) – Only one option possible

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(I) FEATURES & OPTIONS – APU - ENERGY STORAGE APPLICATION

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	COMMENTS
DC precharge circuit	F				
AC precharge circuit	O				
Operating mode: CC-CV	F				
Operating mode: STATCOM	O				
Operating mode: Island grid	O				
Fuse-protected DC Inputs NH 3L	O (1…6) / O (1…4)				coupled / separated (incl. Common Mode Filter) APUs
DC-current monitoring	O				DC input quantity depends on the measuring range
DC disconnector	F				

(O) – Optional (F) – Feature

APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

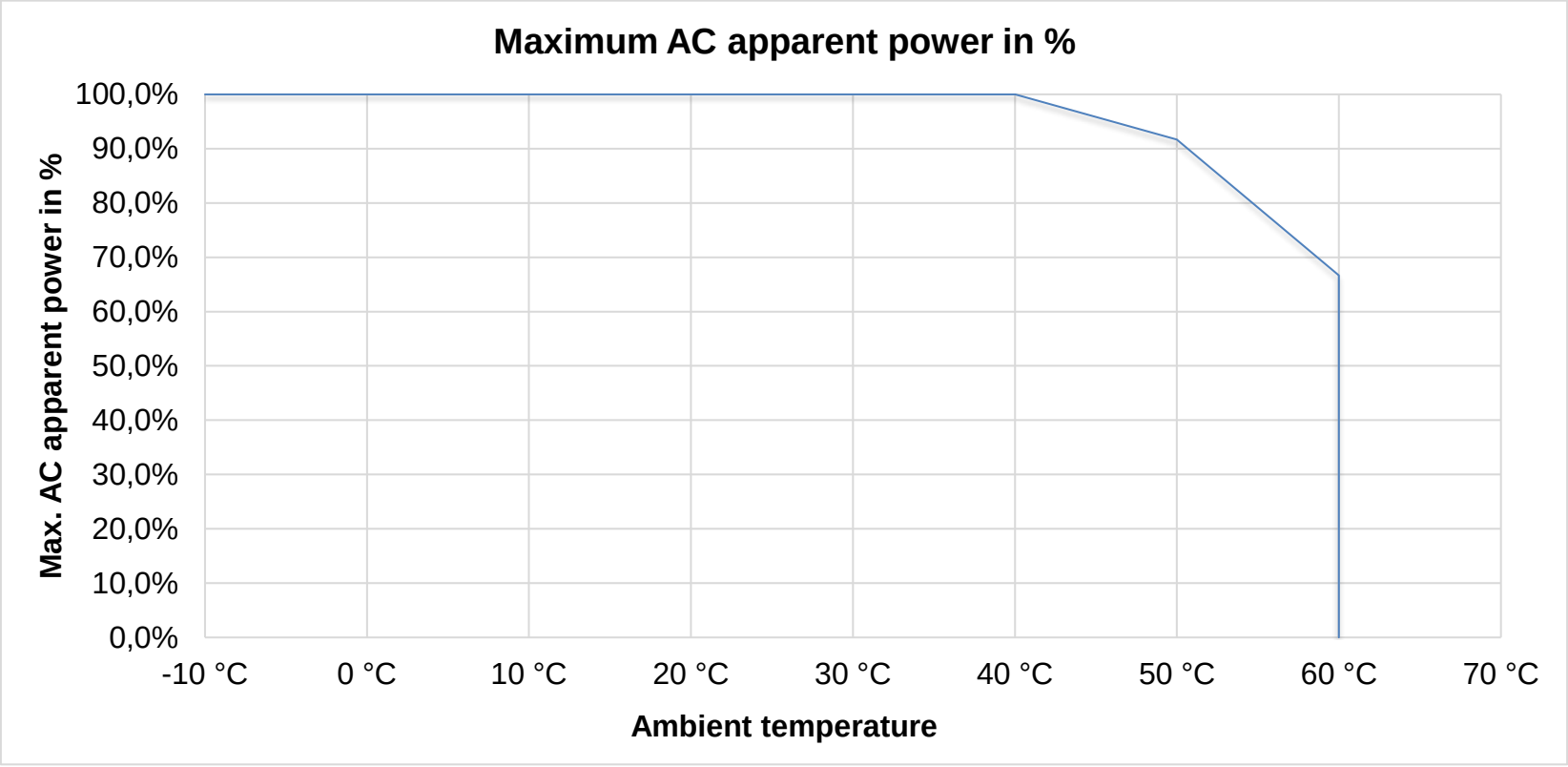
(II) FEATURES & OPTIONS – APU - ENERGY STORAGE APPLICATION

	APS600-ES-1-330-5 APS650-ES-1-360-5 APS725-ES-1-400-5 APS800-ES-1-440-5 APS870-ES-1-480-5 APS945-ES-1-520-5 APS1000-ES-1-550-5 APS1045-ES-1-575-5 APS1090-ES-1-600-5 APS1140-ES-1-630-5 APS1200-ES-1-660-5 APS1250-ES-1-690-5	APS1200-ES-2-330-5 APS1300-ES-2-360-5 APS1450-ES-2-400-5 APS1600-ES-2-440-5 APS1740-ES-2-480-5 APS1890-ES-2-520-5 APS2000-ES-2-550-5 APS2090-ES-2-575-5 APS2180-ES-2-600-5 APS2280-ES-2-630-5 APS2400-ES-2-660-5 APS2500-ES-2-690-5	APS1800-ES-3-330-5 APS1950-ES-3-360-5 APS2175-ES-3-400-5 APS2400-ES-3-440-5 APS2610-ES-3-480-5 APS2835-ES-3-520-5 APS3000-ES-3-550-5 APS3135-ES-3-575-5 APS3270-ES-3-600-5 APS3420-ES-3-630-5 APS3600-ES-3-660-5 APS3750-ES-3-690-5	APS2400-ES-4-330-5 APS2600-ES-4-360-5 APS2900-ES-4-400-5 APS3200-ES-4-440-5 APS3480-ES-4-480-5 APS3780-ES-4-520-5 APS4000-ES-4-550-5 APS4180-ES-4-575-5 APS4360-ES-4-600-5 APS4560-ES-4-630-5 APS4800-ES-4-660-5 APS5000-ES-4-690-5	**COMMENTS**
Monitored surge arresters on DC side	F				
AC circuit-breaker	F				
DC connection	F				APU pairs coupled on DC
	O				APU pairs separated on DC
Island grid preparation	O				
Insulation monitoring	O *)				None
	F *)				Internal
	O *)				External
	O *)				Positive grounding; Short-circuit level: 5 kA
	O *)				Negative grounding; Short-circuit level: 5 kA

(O) – Optional (F) – Feature (*)) – Only one option possible

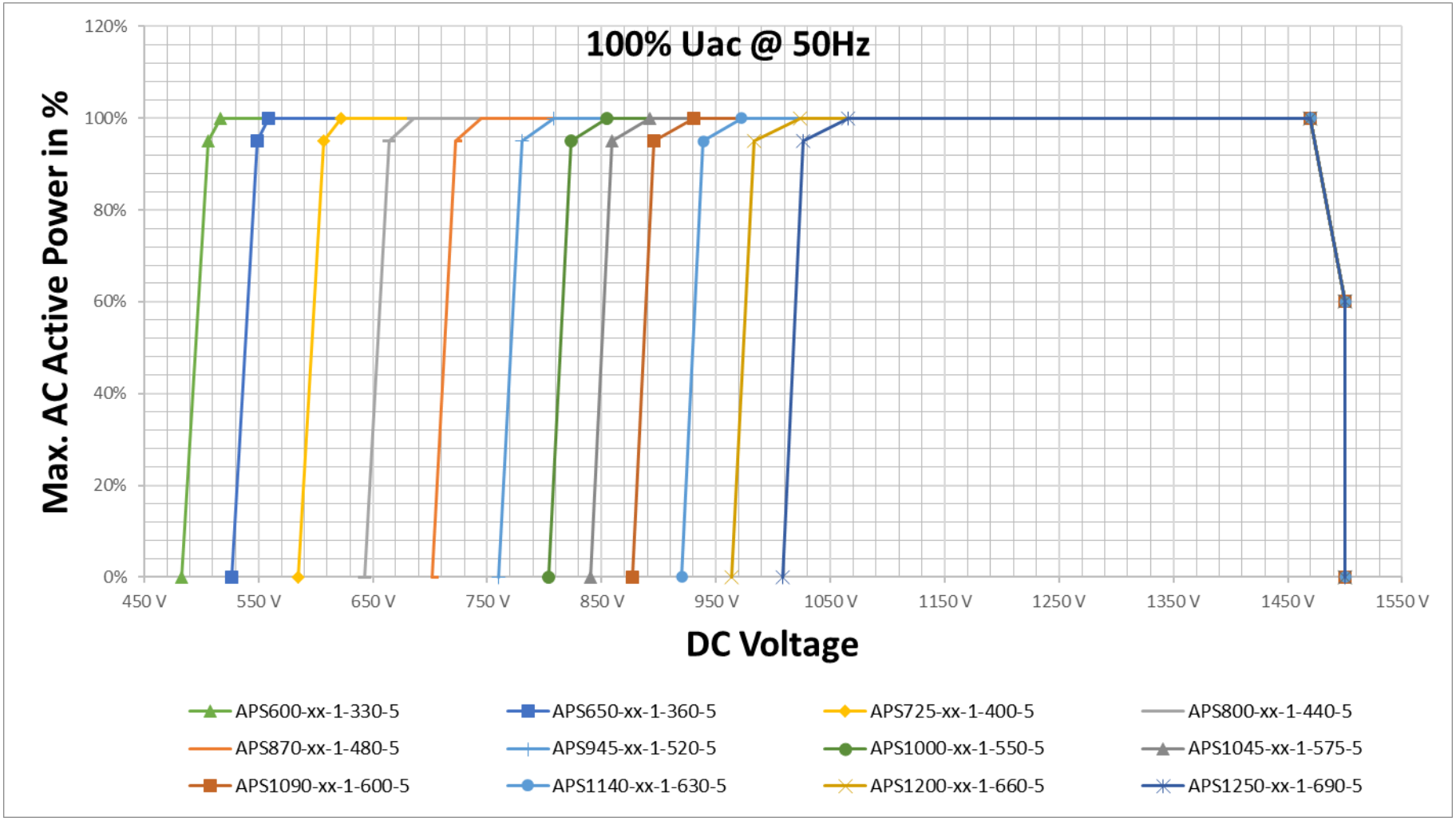
APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

POWER DERATING DEPENDING ON AMBIENT TEMPERATURE



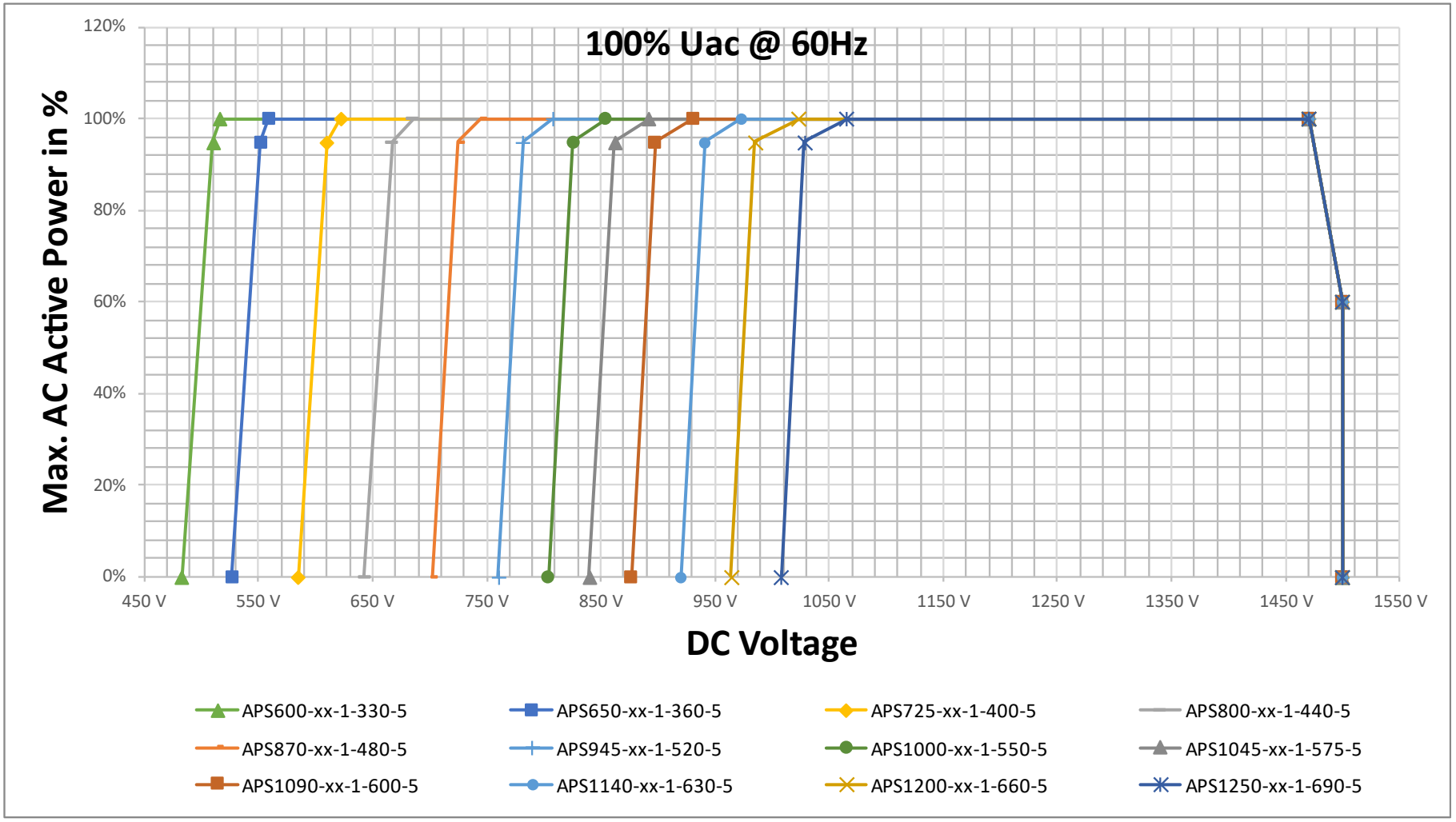
APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

POWER DERATING DEPENDING ON DC VOLTAGE



APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

POWER DERATING DEPENDING ON DC VOLTAGE



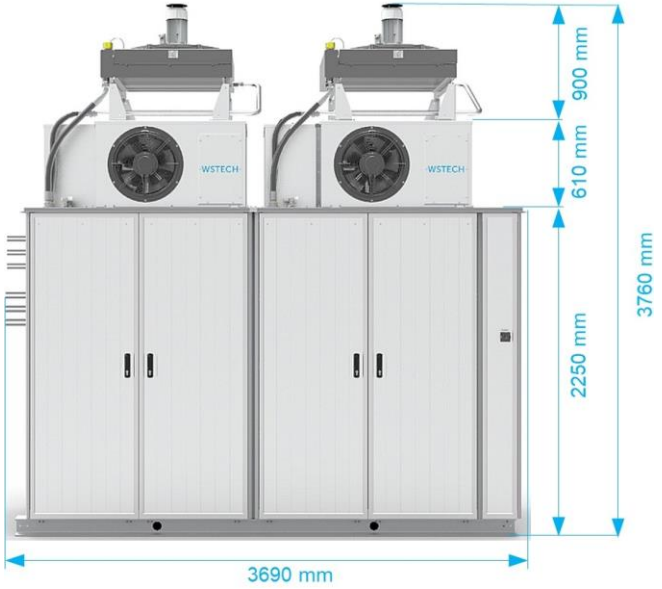
APS-ES-Series 1500 V_{DC} – 330 … 690 V_{AC} 600 … 5000 kVA

(I) MECHANICAL DRAWINGS

Models	**APS600-ES-1-330-5**	**APS1200-ES-2-330-5**	**APS1800-ES-3-330-5**	**APS2400-ES-4-330-5**
	APS650-ES-1-360-5	**APS1300-ES-2-360-5**	**APS1950-ES-3-360-5**	**APS2600-ES-4-360-5**
	APS725-ES-1-400-5	**APS1450-ES-2-400-5**	**APS2175-ES-3-400-5**	**APS2900-ES-4-400-5**
	APS800-ES-1-440-5	**APS1600-ES-2-440-5**	**APS2400-ES-3-440-5**	**APS3200-ES-4-440-5**
	APS870-ES-1-480-5	**APS1740-ES-2-480-5**	**APS2610-ES-3-480-5**	**APS3480-ES-4-480-5**
	APS945-ES-1-520-5	**APS1890-ES-2-520-5**	**APS2835-ES-3-520-5**	**APS3780-ES-4-520-5**
	APS1000-ES-1-550-5	**APS2000-ES-2-550-5**	**APS3000-ES-3-550-5**	**APS4000-ES-4-550-5**
	APS1045-ES-1-575-5	**APS2090-ES-2-575-5**	**APS3135-ES-3-575-5**	**APS4180-ES-4-575-5**
	APS1090-ES-1-600-5	**APS2180-ES-2-600-5**	**APS3270-ES-3-600-5**	**APS4360-ES-4-600-5**
	APS1140-ES-1-630-5	**APS2280-ES-2-630-5**	**APS3420-ES-3-630-5**	**APS4560-ES-4-630-5**
	APS1200-ES-1-660-5	**APS2400-ES-2-660-5**	**APS3600-ES-3-660-5**	**APS4800-ES-4-660-5**
	APS1250-ES-1-690-5	**APS2500-ES-2-690-5**	**APS3750-ES-3-690-5**	**APS5000-ES-4-690-5**







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